

CYBER SECURITY TASK 02

PART A – UNDERSTANDING THE CIA TRIAD

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1. Confidentiality

Definition : Confidentiality is the principle of protecting sensitive information from unauthorized access. It ensures that only authorized users can view or access data.

Why is it Important?

- Protects personal and sensitive information.
- Prevents data theft and unauthorized disclosure.
- Helps organizations maintain customer trust and comply with privacy regulations.

Real-World Example : A college stores student records in a secure database. Only authorized staff members, such as administrators and faculty, can access this information using unique usernames and passwords.

Common Security Measures :

- Strong passwords
- Multi-Factor Authentication (MFA)
- Encryption
- Access control

2.Integrity

Definition : Integrity ensures that data remains accurate, complete, and unchanged unless modified by an authorized person. It protects information from unauthorized alteration or corruption.

Why is it Important?

- Maintains the accuracy of important information.
- Prevents unauthorized changes to data.
- Ensures reliable decision-making based on correct information.

Real-World Example : A university stores students' examination results in its database. Only authorized faculty members are allowed to update grades, ensuring that marks cannot be altered by unauthorized users.

Common Security Measures :

- Hashing
- Digital signatures
- Access permissions
- Audit logs

3.Availability

Definition : Availability ensures that systems, applications, and data are accessible to authorized users whenever they are needed.

Why is it Important?

- Supports uninterrupted business and academic operations.
- Reduces downtime during system failures or cyberattacks.
- Ensures users can access important services when required.

Real-World Example : A college website remains available during the admission period so that students can submit applications, download forms, and access important announcements without interruption.

Common Security Measures :

- Regular backups
- Disaster recovery plans
- Redundant servers
- System monitoring

Comparison Table :

Aspect	Confidentiality	Integrity	Availability
Definition	Protects data from unauthorized access.	Ensures data remains accurate and unaltered.	Ensures systems and data are available when needed.
Primary Goal	Keep information private.	Maintain data accuracy and trustworthiness.	Provide continuous access to systems and information.
Main Risk	Data breaches or information leaks.	Unauthorized modification or corruption of data.	System downtime or service disruption.
Example	Student records accessible only to authorized staff.	Examination results cannot be changed without permission.	College website remains accessible during admissions.
Common Security Measures	Encryption, MFA, Access Control	Hashing, Digital Signatures, Audit Logs	Backups, Redundant Servers, Disaster Recovery